

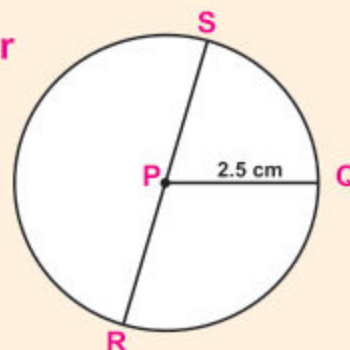
**Activity 2**

Draw a circle using ruler whose radius is 2.5 cm.

$$m \overline{PQ} = \text{radius} = 2.5 \text{ cm}$$

$$m \overline{PQ} = 2.5 \text{ cm}$$

$$\begin{aligned} m \overline{SR} &= \text{diameter} = \text{twice the radius} \\ &= 2 \times 2.5 = 5.0 \text{ cm} = 5 \text{ cm} \end{aligned}$$

**EXERCISE 6.12**

By using compass and ruler draw the following circles if their radii are:

(1) 3.2 cm

(2) 4.1 cm

(3) 4.4 cm

(4) 5.5 cm

(5) 6 cm

(6) 4.8 cm

6.5 QUADRILATERALS

Construct squares and rectangles with sides of given measure using protractor, set squares and straightedge/ruler.

**Activity 1**

Draw a square with side 4 cm using set square.

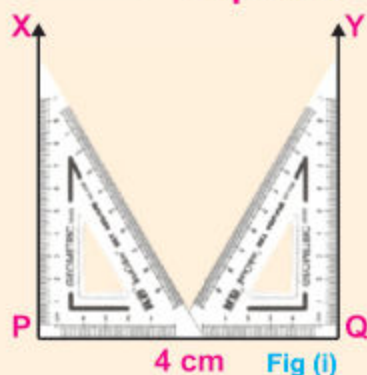


Fig (i)

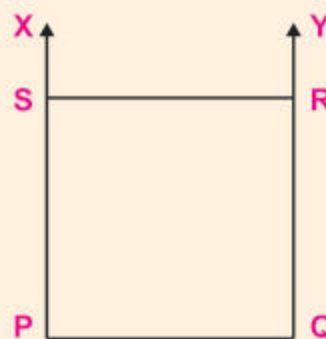


Fig (ii)

Steps of construction:

Step 1. Draw $\overline{PQ}$, 4 cm long.

Step 2. At points P and Q, draw right angles $\angle QPX$ and $\angle PQY$ using set square. [see fig (i)]

Step 3. From $\overrightarrow{PX}$, measure $\overline{PS} = 4 \text{ cm}$ and from $\overrightarrow{QY}$, measure $\overline{QR} = 4 \text{ cm}$. Join R and S. So, PQRS is the required square [see fig (ii)].

**Activity 2**

To draw a rectangle with sides 6.2 cm and 3.5 cm using protractor and ruler.



Step 1. Draw $\overline{AB}$, 6.2 cm long.

Step 2. At points A and B draw right angles $\angle BAE$ and $\angle ABF$ using protractor.

Step 3. From $\overrightarrow{AE}$, measure $\overline{AD} = 3.5$ cm and from $\overrightarrow{BF}$, measure $\overline{BC} = 3.5$ cm.

Step 4. Join C and D. So, ABCD is the required rectangle.

EXERCISE 6.13

1. Draw squares with sides of the following lengths. Using protractor or set square.

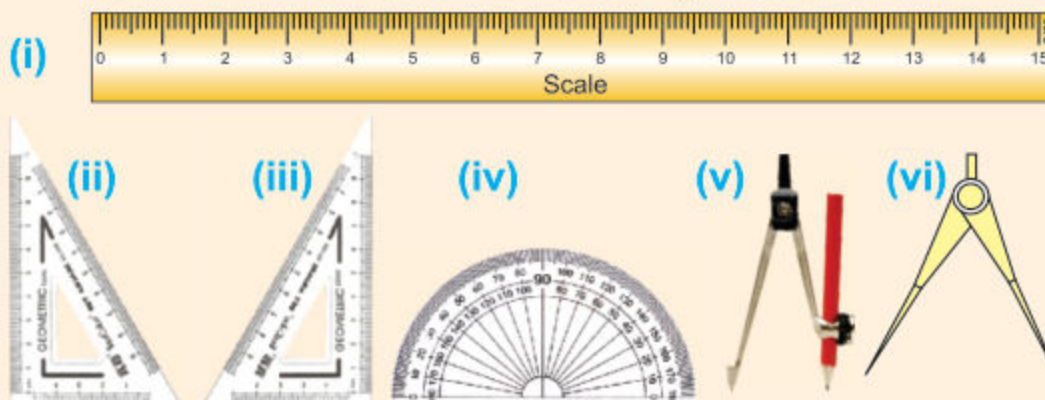
(i) 6.5 cm (ii) 4.9 cm (iii) 5.8 cm (iv) 6.3 cm
(v) 7.3 cm (vi) 4.1 cm (vii) 8.7 cm (viii) 8.5 cm

2. Draw rectangle with sides of the following lengths using protractor or set square.

(i) 8 cm and 7 cm (ii) 9.5 cm and 4.7 cm
(iii) 12.3 cm and 5.2 cm (iv) 9.6 cm and 3.7 cm
(v) 6.1 cm and 2.9 cm (vi) 2.9 cm and 7.6 cm

REVIEW EXERCISE – 6

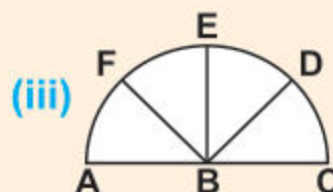
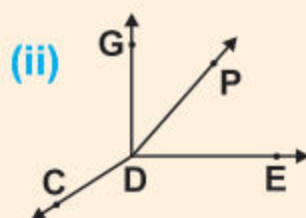
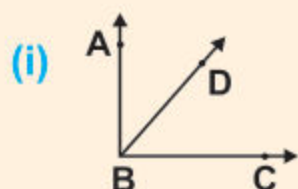
1. Write down the names of the following instruments.



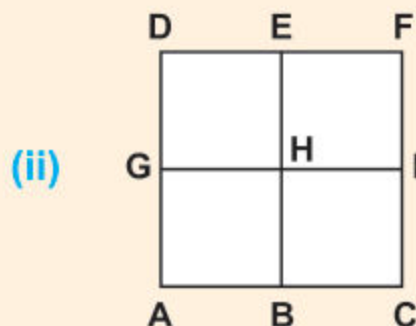
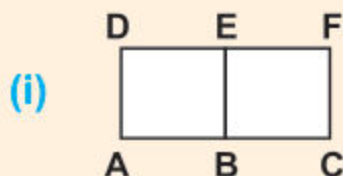
2. If you have to draw a thin line segment, which grade of pencil you will use.
3. Measure the length and width of the following objects with ruler in millimetres.
- (i) A post card (ii) Cover of Maths-book
(iii) Geometry Box
4. Draw a pair of :
- (a) Parallel and non-parallel lines.
(b) Horizontal and vertical lines.
5. Draw a curved line and measure its length by using thread and ruler.
6. Draw line which passes through a given point and is parallel to a given line using set square.
7. Draw the figure to go with each sentence:
- (i) $\overleftrightarrow{PQ}$ is parallel to $\overleftrightarrow{RS}$.
(ii) $\overleftrightarrow{UV}$ intersects $\overleftrightarrow{PQ}$ at point A.
(iii) $\overleftrightarrow{UV}$ intersects $\overleftrightarrow{RS}$ at point B.

8. List some objects in your class room that represent right angles.

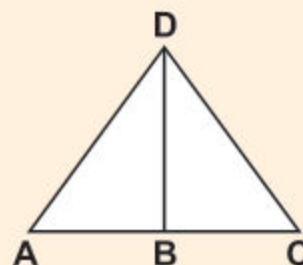
9. How many angles can you find in the figure?
Name and mention the type of the angle.



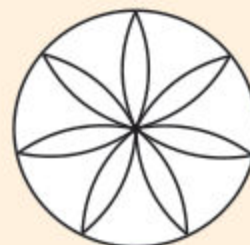
10. Count the number of squares in the figure. Also name them.



11. How many triangles in the figure?



12. Get a compass and practice drawing same circles. Make Some designs of your own.



13. (i) Draw a circle with 25 mm radius.
(ii) Name its centre, radius, diameter and circumference.
(iii) Measure its circumference with thread and ruler.

7.1 BAR GRAPH.

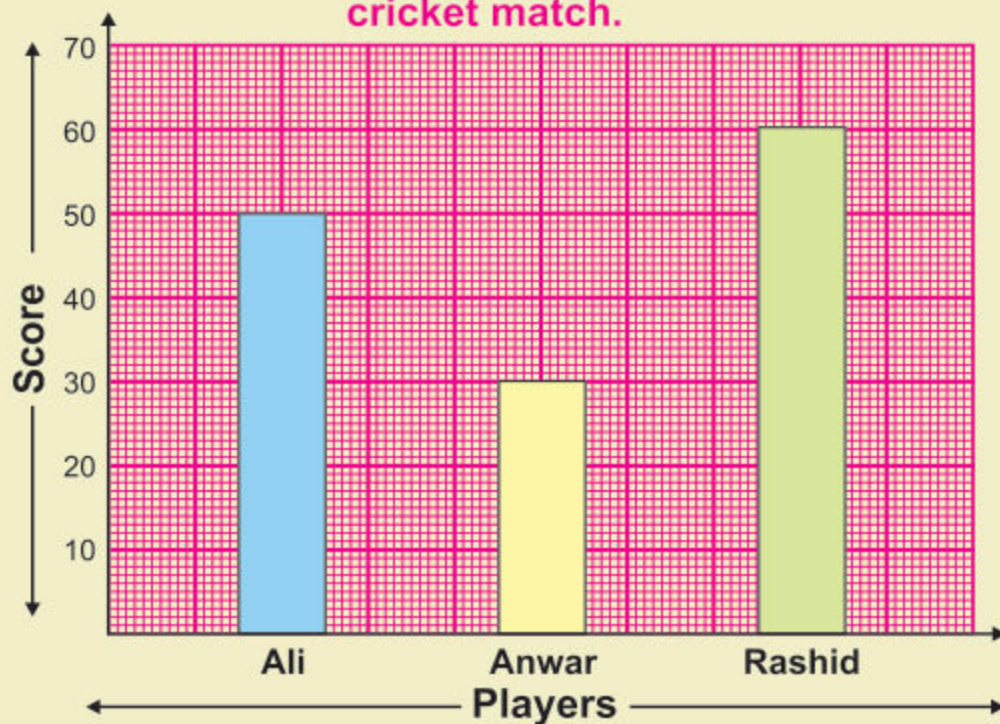
Read and interpret simple bar graphs given in horizontal and vertical form.

In bar graph we draw rectangular strips, horizontally or vertically with equal spacing between them. Each bar represents only one quantity.



Activity

Following is the picture of a bar graph showing the score of three players in a cricket match.



From the above bar graph. We can understand the following.

The number of runs made by Ali are **50**.

The number of runs made by Anwar are _____.

The number of runs made by Rashid _____.

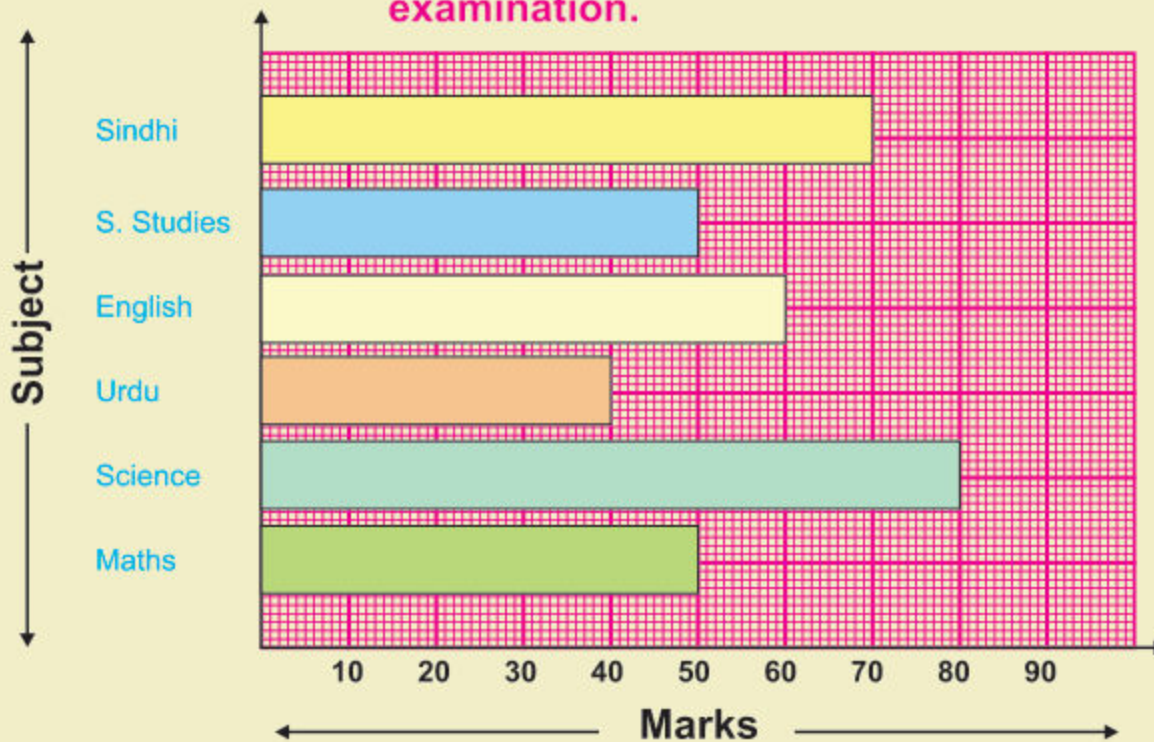
Whose runs are the highest _____?

Whose runs are the least _____?



Activity

Read the bar graph showing the marks secured by Saima in her final examination.



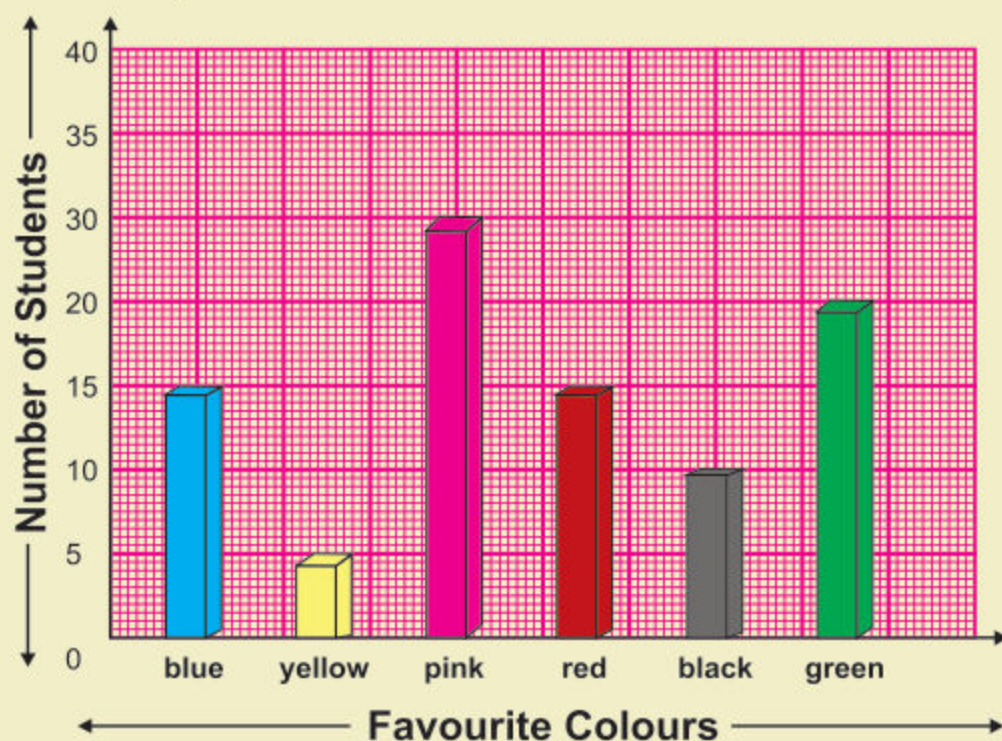
Fill in the following boxes:

- 1 She secured highest marks in:
- 2 She secured lowest marks in:
- 3 In how many subjects she appeared?
- 4 In how many subjects she secured equal marks?
- 5 In which subject she secured more than 60 marks?
- 6 What was the total marks of Saima?

EXERCISE 7.1

1 Look at this Bar graph.

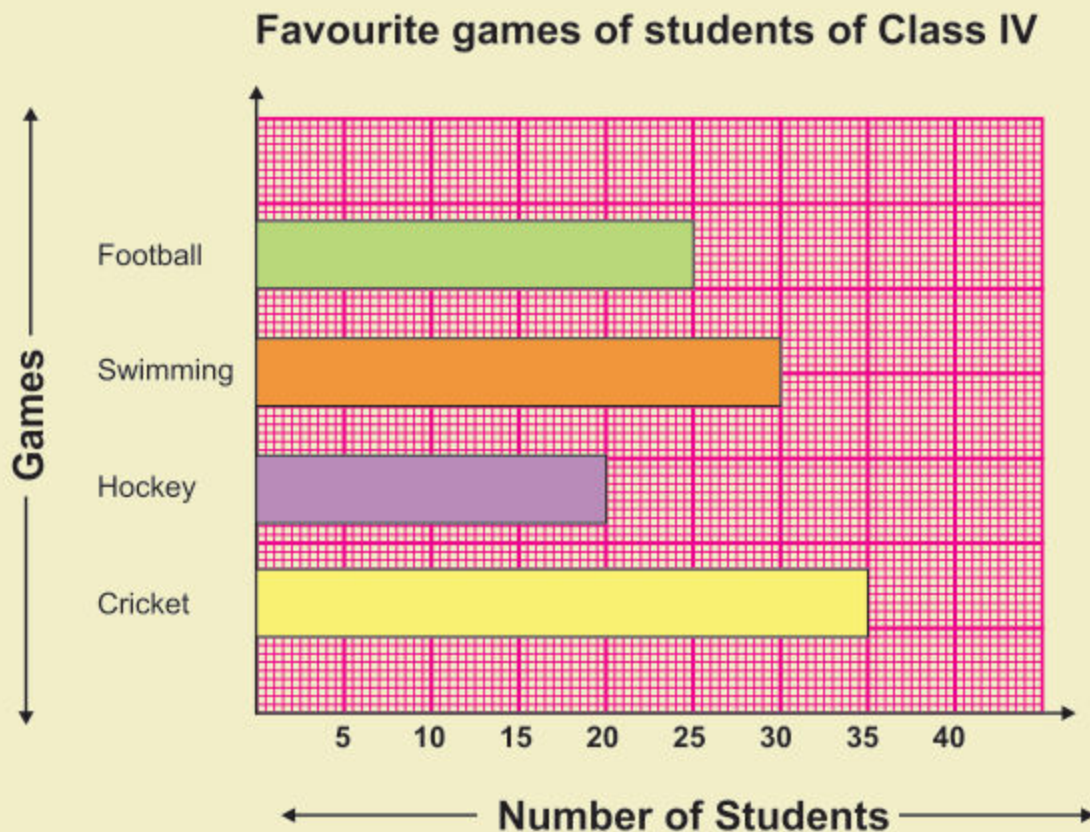
Survey of favourite colours of students of class IV



Answer the following questions:

- (i) How many students like yellow colour?
- (ii) Which colour is liked most?
- (iii) Which colour is liked least?
- (iv) How many students like green colour?
- (v) How many students like red colour?
- (vi) How many colours are included in this survey?
- (vii) How many students are included in this survey?

2 Read the following bar graph and answer the questions.



- (i) How many students like Hockey?
- (ii) Which game is liked most?
- (iii) How many students like Football?
- (iv) How many games liked by students?
- (v) Which game is liked least?
- (vi) How many students like cricket?
- (vii) How many total students involved in games?

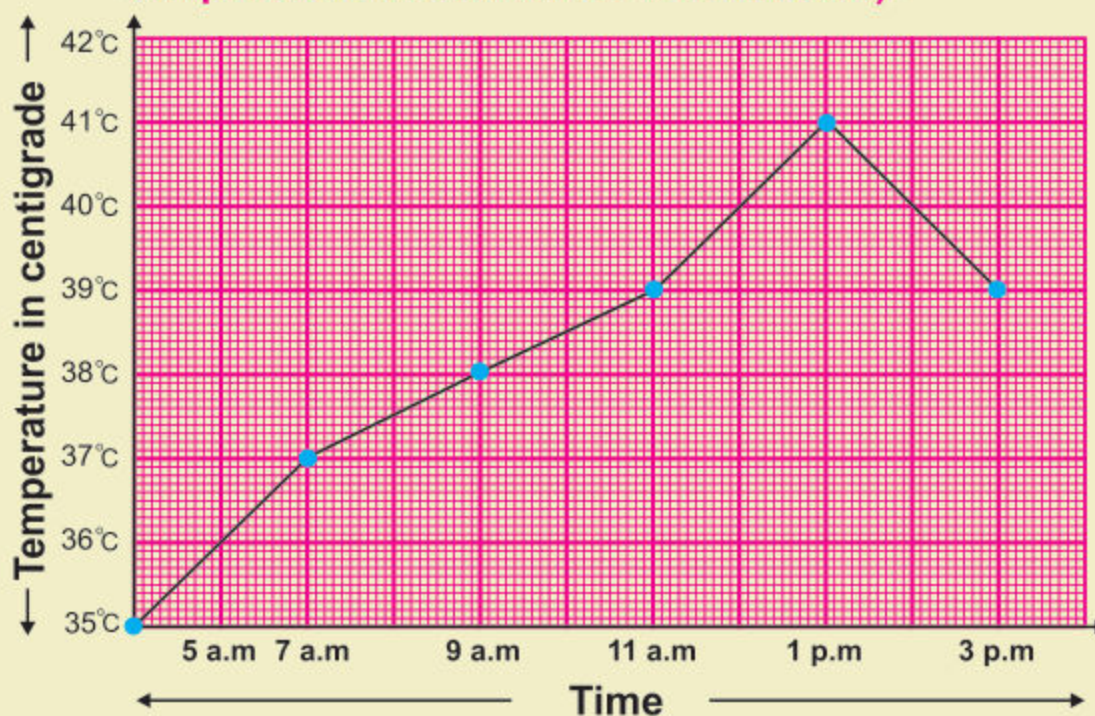
7.2 LINE GRAPH:

Read and interpret line graph

In line graph the information is represented in the form of points. These points are joined together by line segments.



Activity Following is a line graph representing temperature of Jacobabad town at various time of the day. (Time is on horizontal axis and temperature is shown on vertical axis).



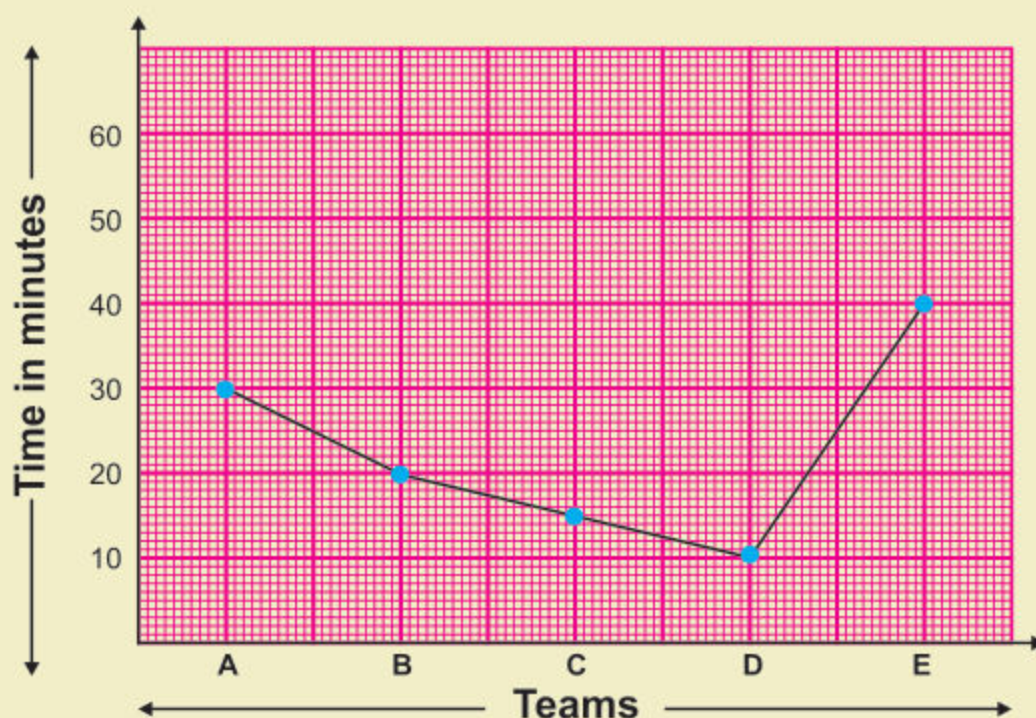
Look at the graph and answer the following questions.

- 1 What was the temperature at 9 a.m.?
- 2 At what time the temperature was the lowest?
- 3 Were the temperatures same at 7 a.m. and 3 p.m.?
- 4 What was the temperature at 1 p.m.?
- 5 What was the lowest temperature?

EXERCISE 7.2

- 1 This line graph shows the time in minutes taken by each of the five teams to complete a task.

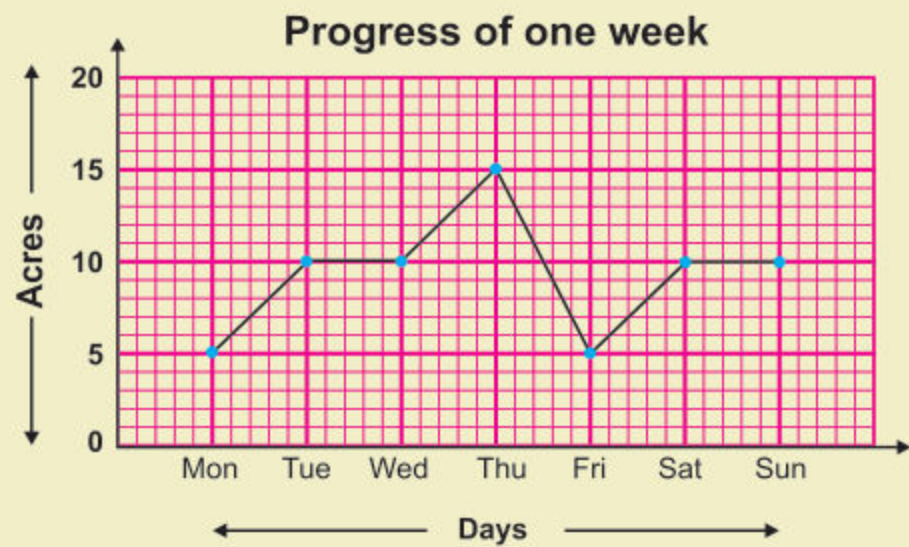
Time Table by Teams to complete the task



Now answer these questions.

- (i) Which team completed the work in the shortest time?
- (ii) Which team took the longest time?
- (iii) Which teams took the same length of time?
- (iv) What was the shortest time?
- (v) What was the longest time?
- (vi) How long did team C take to complete the task?

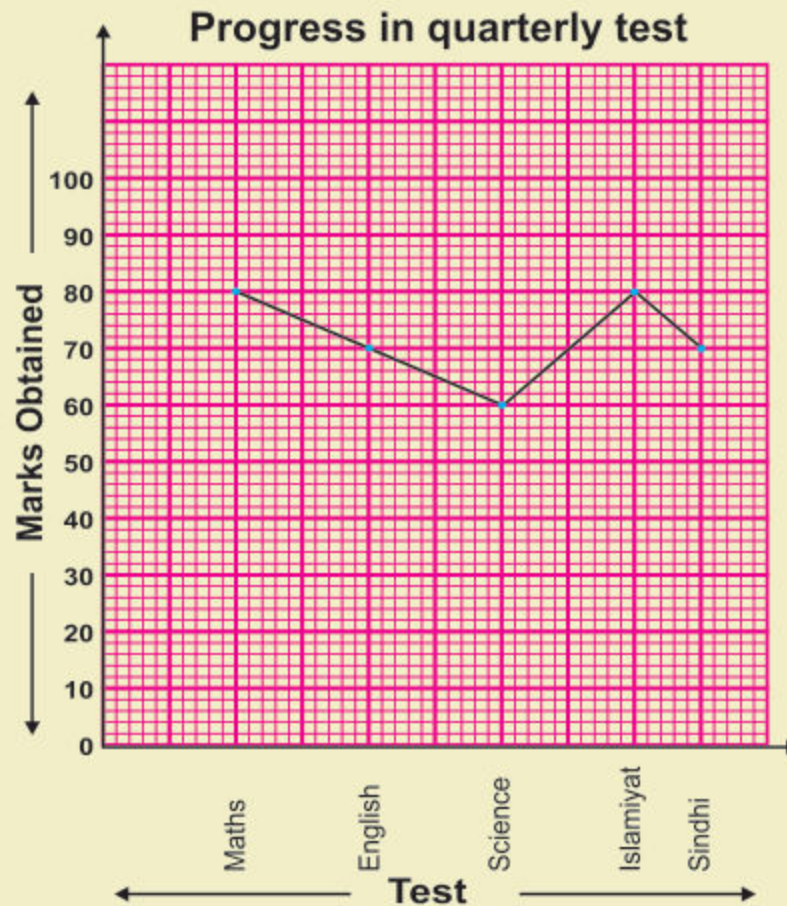
- 2 A farmer took a tractor on rent to plough his fields. The progress of one week is shown by the following graph. Names of days are shown on horizontal axis and acres on vertical axis.



Look at the graph and answer the following questions:

- (i) On which day the tractor ploughed the maximum
- (ii) How many acres of land were ploughed on
- (iii) For how many days the land was ploughed?
- (iv) What was his progress on Tuesday?
- (v) On which days the progress was minimum?
- (vi) How many acres of fields were ploughed during the whole week?
- (vii) Give the separate output for ,
- (viii) On which days the output was same?
-

- 3 Samina's progress in quarterly tests is shown in the graph. The names of subjects are shown on horizontal axis. The number of marks are shown on vertical axis.

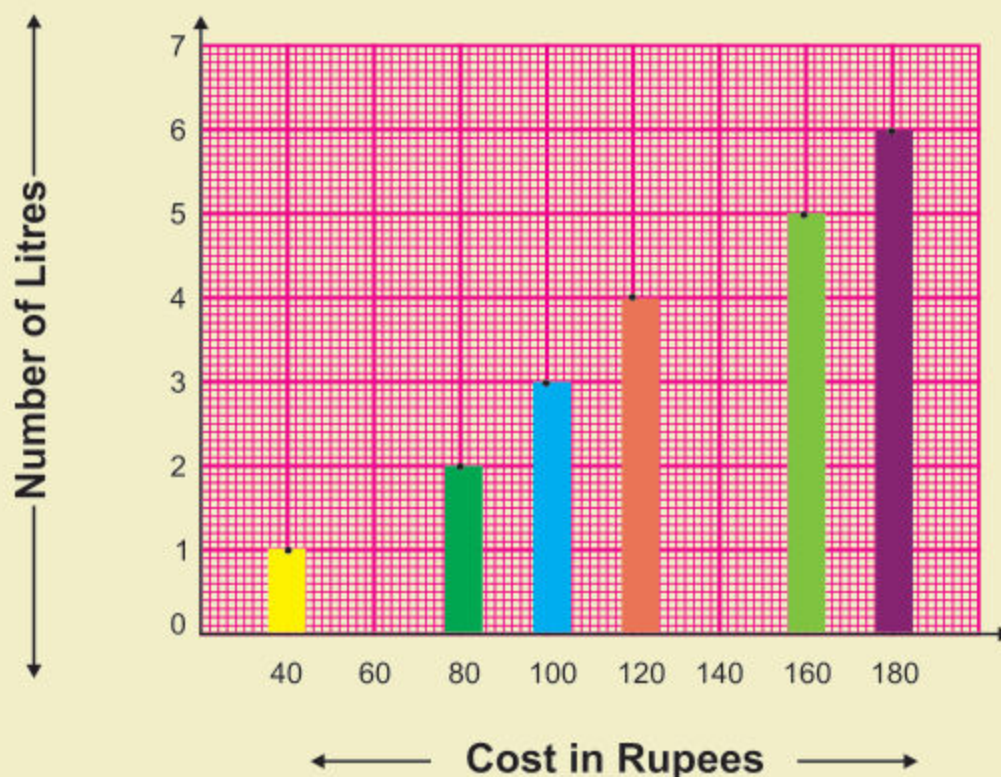


Look at the graph and answer the following questions:

- (i) In which subject(s) Samina secured the highest marks?
- (ii) What were her total marks obtained?
- (iii) How many marks did she get in English?
- (iv) How many marks did she get in Maths?
- (v) In which subject did she get the lowest marks?
- (vi) How many marks did she get in Sindhi?

REVIEW EXERCISE – 7

(1) The given bar graph shows the cost of mango squash.



Study the graph and answer these questions.

(i) What is the cost of 2 litres of mango squash?

(ii) What is the cost of 4 litres of mango squash?

(iii) If we have six notes of Rs 20, can we buy 6 litres of mango squash?

(iv) How much change, we will have from Rs 100, when we buy 2 litres of mango squash?

(v) How much litres of squash can we buy from Rs 180

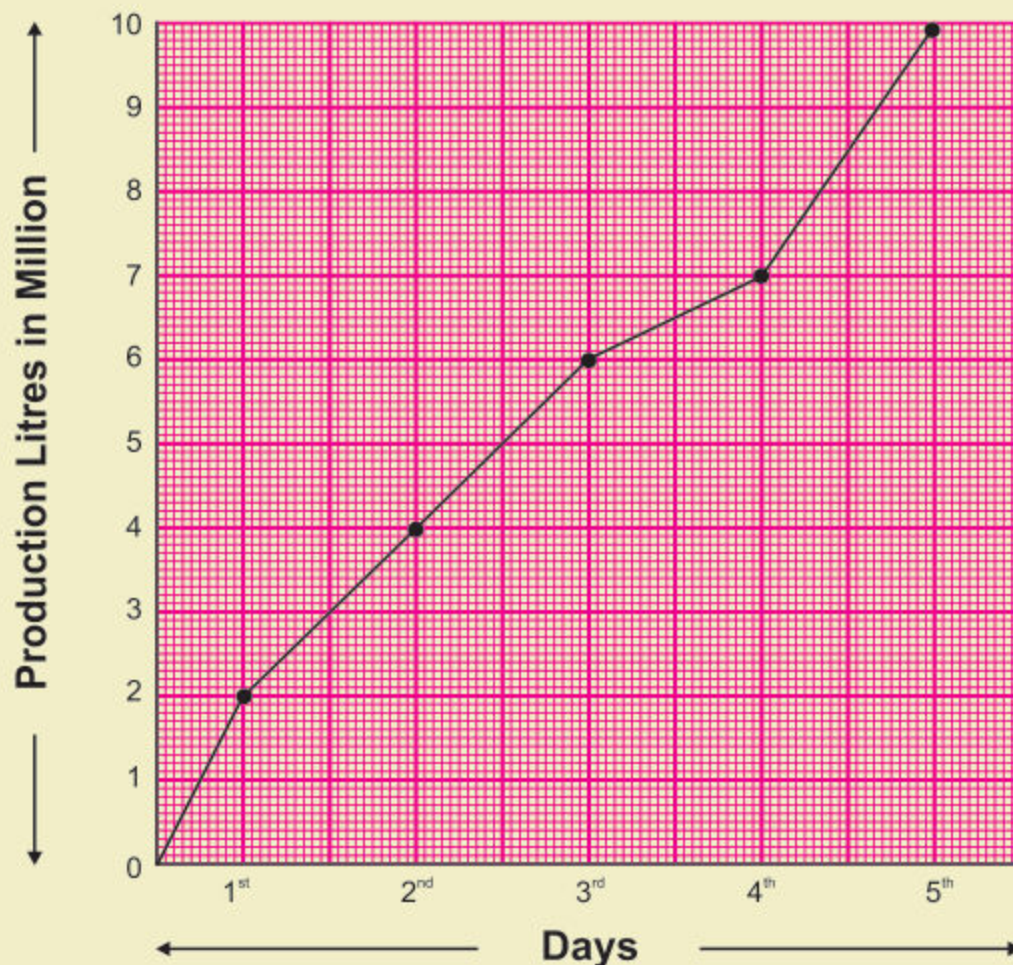
- (2) Read the following bar graph of Rashid's family about the expenditure on food for six months.



Look at the graph and answer the following questions.

- (i) How much amount is spent on food during the month of February?
- (ii) How much amount is spent on food during month of April?
- (iii) In which month the expenditure spent was the lowest?
- (iv) In which month the expenditure spent was the highest?
- (v) What are the total expenditures spent from January to June?
- (vi) What was the difference in expenditure spent for April and May.

- (3) The line graph shows the production of a oil plant over a period of 5 days.



Look at the line graph and answer these questions.

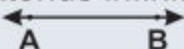
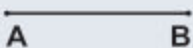
- (i) What was the production of the plant on the 1st day?
- (ii) What was its production on the 3rd day?
- (iii) What production was made between the 2nd and 3rd day?
- (iv) How much production was made between the 2nd and 5th days?
- (v) On which day the plant growth production was the highest?
- (vi) How many times the plant production was measured?

GLOSSARY

Addition:	Symbol +; the process of finding sum of two numbers/quantities.
Associative property of addition:	When any three numbers or fractions added in any order, their sum is always the same.
Associative property of multiplication:	When any three numbers or fractions are multiplied in any order, their product is always the same.
Angle:	The amount of turning between two arms about a common point.
Arc:	A part of the circumference of a circle.
Acute angle:	An angle of measure greater than 0° and less than 90° .
Acute angle triangle:	A triangle which has three acute angles.
Capacity:	The amount of liquid a container can hold.
Centimetre:	A unit of length, 100 centimetres (cm) = 1 metre (m)
Circle:	The set of all the points which are equidistant from a fixed point.
Commutative property of Multiplication:	Any two numbers or fractions when multiplied to each other in any order, their product is always same.
Commutative property of addition:	The property that when any two numbers (fractions) are added in any order their sum is always same.
Common multiples:	The numbers which are common in multiples of two or more numbers.
Composite number:	A natural number which has more than two factors.
Division:	Process of finding the quotient of two numbers/quantities. (The repeated subtraction).
Denominator:	Lower part of the fraction.
Diameter:	A line segment that divides the circle into two equal halves



GLOSSARY

Divisibility:	When a number is divided exactly by another number, (the remainder is zero).
Dividend:	A number is to be divided by another number.
Divisor:	A number by which another number is to be divided.
Decimal fraction:	A fraction whose denominator is a power of ten and whose numerator is expressed by figures placed to the right of a decimal point.
Even numbers:	A number that can be divided into two equal groups. Even numbers end in 2,4,6,8 and 0 regardless of how many digits they have.
Edge:	The outside limit of an object, area, or surface.
Equivalent fraction:	Fractions with different numerators and denominators that represent the same value.
Fraction:	Part of a whole.
Factors:	Factors are whole numbers that are multiplied together to produce another number.
Factorization:	a method of writing numbers as the product of their factors or divisors.
Gram:	a metric unit of mass equal to one thousandth of a kilogram.
Graph:	A pictorial representation of data.
GCD:	Greatest Common Divisor.
Hours:	A unit of time 1 hour = 60 minutes
HCF:	Highest Common Factor.
Improper fraction:	A fraction whose numerator is greater than or equal to denominator.
Kilogram:	A unit of mass. 1 kilogram (kg) = 1000 grams (g)
Litre:	Unit of volume/capacity 1 litre (l) = 1000 millilitres (ml)
Line:	A line of points that extends infinitely in two directions 
Line segment:	A line with two definite end points 

GLOSSARY

Lunar Calendar:	(Hijrah Qamri Calendar) Islamic Calendar in a solar year.
L.C.M	Lowest Common Multiple.
Like fractions:	Fractions having same denominator.
Multiplication:	The process of finding product of two numbers/quantities (Repeated Addition).
Mass:	Mass is commonly measured by how much something weighs.
Millilitre:	Thousandth part of a litre.
Millimetre:	Thousandths part of a metre.
Minute:	A unit of time. 1 minute = 60 seconds
Month:	A unit of time. 1 month = 30 days
Million:	The smallest seven digit number i.e. 1,000,000
Mixed number:	a number consisting of a whole number and a proper fraction.
Numerator:	Upper part of a fraction.
Obtuse angle:	An angle of measure greater than 90° and less than 180° .
Obtuse angled triangle:	A triangle with one obtuse angle and two acute angles.
Place value:	Value of a digit according to its position in a number
Proper fraction:	a fraction that is less than one, with the numerator is less than the denominator.
Paisa:	Unit of Pakistani currency.
Point:	A point is a location. It has no size i.e. no width, no length and no depth. It is shown by a dot.
Prime factorisation	A factorization in which every factor is a prime number
Protector:	An instrument for measuring angles.
Quadrilateral:	A four sided closed figure.
Quotient:	A result obtained by dividing one quantity by another. The number shows how many times the divisor has been repeatedly subtracted.

GLOSSARY

Remainder:	The number left over when one integer is divided by another.
Ray:	An arrow mark on one end point of a line segment 
Rectangle:	A quadrilateral whose opposite sides are equal and have four right angles.
Radius:	The distance from centre to any point on the circumference of circle.
Rupee:	Unit of Pakistani currency.
Ruler:	A straight edge instrument used to measure distances.
Right angle:	An angle of measure 90° .
Right triangle:	A triangle in which one angle is of right angle.
Symbol:	A sign used to represent an operation, element or relation.
Square:	A quadrilateral with four equal sides and four right angles.
Subtraction:	The process of taking one number or amount away from another
Second:	Unit of time, $1 \text{ second} = \frac{1}{60} \text{ minute}$.
Solar Calendar:	In this calendar, the dates indicates the position of earth around the sun (365 days in a year).
Scalene triangle:	A triangle whose all sides and angles are of different measure.
Triangle:	A three sided closed geometrical figure.
Unlike fractions:	The fractions with different denominators.
Unit fraction:	A proper fraction whose numerator is equal to 1.
Vertex:	An angular point of any shape.
Week:	A unit of time. 1 week = 7 days
Year:	A unit of time. 1 year = 365 days

ANSWERS

EXERCISE 1.1

- (1) (i) 4,692 (ii) 69,407 (iii) 456,926 (iv) 700,000
 (v) 902,042 (vi) 729,006 (vii) 902,042 (vi) 729,006
- (2) (i) Two thousand five hundred sixty one
 (ii) Thirty four thousand five hundred sixty one
 (iii) Fifty six thousand eighty one (iv) Ninety two thousand
 (v) Two hundred forty five thousand six hundred twelve
 (vi) Three hundred forty nine thousand six hundred fifty only

EXERCISE 1.2

- (1) (i) The place value of 2 is 2 million = 2,000,000
 (ii) The place value of 8 is 8 million = 8,000,000
 (iii) The place value of 1 is 1 hundred million = 100,000,000
 (iv) The place value of 9 is 9 hundred thousand = 900,000
 (v) 5 million = 5,000,000, (vi) 9 million = 9,000,000,
 2 hundred = 200 1 thousand = 1000
 (vii) 9 hundred thousand = 900,000, (viii) 7 ten million = 70,000,000,
 8 thousand = 8000 6 million = 6,000,000
- (2) (i) 9 millions = 9,000,000
 2 hundred thousands = 200,000
 3 ten thousands = 30,000
 4 thousands = 4,000
 5 hundreds = 500
 1 ten = 10
 3 ones = 3
- (ii) 5 ten millions = 50,000,000
 0 millions = 0,000,000
 1 hundred thousands = 100,000
 2 ten thousands = 20,000
 0 thousand = 0,000
 3 hundreds = 300
 0 tens = 00
 6 ones = 6
- (iii) 3 millions = 3,000,000
 5 hundred thousands = 500,000
 6 ten thousands = 60,000
 7 thousands = 7,000
 8 hundreds = 800
 9 tens = 90
 9 ones = 9
- (iv) 3 ten millions = 30,000,000
 6 millions = 6,000,000
 5 hundred thousands = 500,000
 6 ten thousands = 60,000
 4 thousands = 4000
 3 hundreds = 300
 9 tens = 90
 6 ones = 6

ANSWERS

EXERCISE 1.3

1.
 - (i) Two hundred forty one thousand and nine hundred thirty five.
 - (ii) Four million, three hundred twelve thousand and six hundred eighty seven.
 - (iii) Five million.
 - (iv) Twenty five million, one hundred thirty four thousand and five hundred sixty four.
 - (v) One hundred million.
 - (vi) Nine million, two hundred sixty four thousand and three hundred eighty seven.
 - (vii) Fifty millions one thousand.
 - (viii) Four million, one hundred nine thousand and two hundred.
 - (ix) Ninety nine million, nine hundred ninety thousand and ninety.
2.
 - (i) 2,900,068 (ii) 30,600,045 (iii) 100,000,000
 - (iv) 20,000,020 (v) 90,067,000
4.
 - (i) 2,453,761, 2,454,761, 2,455,761
 - (ii) 7,000,600, 7,000,700, 7,000,800
 - (iii) 67,243,415, 67,253,415, 67,263,415

5.

Millions	Ten Millions	Hundred Millions
2,456,178	22,233,341	100,000,000
1,000,000	10,000,000	
2,561,000	20,001,010	

EXERCISE 1.4

1.
 - (i) > (ii) < (iii) < (iv) > (v) >
2.
 - (i) **Ascending Order:**
3,076,005, 3,174,215, 3,741,512
Descending Order:
3,741,512, 3,174,215, 3,076,005
 - (ii) **Ascending Order:**
95,113,417, 95,123,415, 95,312,415
Descending Order:
95,312,415, 95,123,415, 95,113,417
 - (iii) **Ascending Order:**
52,111,222, 58,110,176, 59,178,215, 59,296,712
Descending Order:
59,296,712, 59,178,215, 58,110,176, 52,111,222

A N S W E R S

(iv) **Ascending Order:**

13,100,219, 13,200,415, 14,111,920, 14,160,000

Descending Order:

14,160,000, 14,111,920, 13,200,415, 13,100,219

EXERCISE 1.5

- | | | | | |
|----|---------------------------|-----------------------------|--------------------------------|--------------------------------|
| 1. | (i) 66676
(v) 1393289 | (ii) 142523
(vi) 1311571 | (iii) 405253 | (iv) 551177 |
| 2. | (i) 114552
(v) 1138719 | (ii) 317981
(vi) 1334462 | (iii) 1114946
(vii) 1145991 | (iv) 1533612
(viii) 1791275 |

EXERCISE 1.6

- | | | | | | |
|----|--------------|----|------------------|----|----------------|
| 1. | Rs 928117 | 2. | 47842 students | 3. | Rs 1498800 |
| 4. | 98682 people | 5. | 1590330 bicycles | 6. | 796310 Mangoes |

EXERCISE 1.7

- | | | | | |
|----|-------------------------------------|---|--------------------------------------|--|
| 1. | (i) 12111
(v) 161547 | (ii) 562051
(vi) 46952 | (iii) 611620 | (iv) 608819 |
| 2. | (i) 55555
(v) 188750
(ix) 999 | (ii) 234122
(vi) 30283
(x) 581110 | (iii) 1
(vii) 90100
(xi) 37908 | (iv) 162136
(viii) 100000
(xii) 449008 |

EXERCISE 1.8

- | | | | | | |
|----|-----------|----|----------------|----|------------|
| 1. | Rs 445550 | 2. | 14001 students | 3. | Rs 272920 |
| 4. | Rs 180002 | 5. | 112 people | 6. | 46240 hens |

EXERCISE 1.9

- | | | | | |
|----|----------------------------|------------------------------|----------------------------------|---------------------------------|
| 1. | (i) 37536
(v) 28880618 | (ii) 163870
(vi) 31791975 | (iii) 328328
(vii) 20578868 | (iv) 177156
(viii) 12087428 |
| 2. | (i) 1168900
(v) 8338834 | (ii) 2579880
(vi) 1518435 | (iii) 14212653
(vii) 17437041 | (iv) 4886040
(viii) 35974913 |

ANSWERS

EXERCISE 1.10

- | | | |
|-------------------|--------------------|----------------|
| 1. 702768 bottles | 2. Rs 756000 | 3. Rs 11635000 |
| 4. 17316000 eggs | 5. 40083820 litres | 6. Rs 9268400 |

EXERCISE 1.11

- | | | | |
|-------------|----------|-----------|----------|
| (1) (i) 562 | (ii) 253 | (iii) 561 | (iv) 782 |
| (v) 123 | (vi) 452 | | |
- (2) (i) Quotient = 156 (ii) Quotient = 254 (iii) Quotient = 158
 (iv) Quotient = 252 (v) Quotient = 245 (vi) Quotient = 369
 (3) Quotient = 145, Remainder = 0 (4) Quotient = 356, Remainder = 0
 (5) Quotient = 2586, Remainder = 3 (6) Quotient = 617, Remainder = 12

EXERCISE 1.12

- | | | |
|---------------|-------------|----------------|
| (1) 896 suits | (2) Rs. 586 | (3) Rs. 225 |
| (4) 281 kg | (5) Rs. 123 | (6) 156 crates |

EXERCISE 1.13

- | | | | |
|---------|----------|----------|------------|
| (1) 40 | (2) 55 | (3) 76 | (4) 58 |
| (5) 354 | (6) 411 | (7) 852 | (8) 490 |
| (9) 115 | (10) 288 | (11) 248 | (12) 175 |
| (13) 82 | (14) 103 | (15) 823 | (16) 10358 |

EXERCISE 1.14

- | | | |
|-----------------|--------------------|------------------------------|
| (1) Rs 1378640 | (2) Rs 205501 | (3) 210320 chicken |
| (4) Rs 14310000 | (5) 47028035 trees | (6) Rs 125 (7) 14 food packs |

REVIEW EXERCISE 1

- (1) (i) c (ii) b (iii) d
- (2) (i) Two million, four hundred twelve thousand and three hundred sixteen
 (ii) Thirty six million, one hundred twenty three thousand and one hundred one
 (iii) Six hundred thousand and two hundred sixteen
- (3) (i) 20,000,000 (ii) 100,000,000

ANSWERS

- (4) (i) 1128857 (ii) 1712258
 (5) (i) 749007 (ii) 100091 (iii) 167778
 (6) (i) 15984423 (ii) 2590560 (iii) 254 (iv) 159 (v) 74
 (7) The amount left with Raheel = Rs 54560

EXERCISE 2.1

- (1) (i), (iii) and (v) (2) (i), (iii) and (v) (3) (i), (iii) and (v)
 (4) (i), (iii) and (v) (5) (i), (iii) and (v)

EXERCISE 2.2

- (1) (ii), (vii) and (ix) (2) (i), (iv), (v), (viii), (ix) and (x)
 (3) **Prime Numbers:** 41, 43, 47, 53, 59. Remaining are composite.
 (4) 2, 3, 5, 7, 11, 13, 17, 19
 (5) 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28
 (6) **Prime Numbers:** 23, 29, 31, 37

Composite numbers: 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39.

EXERCISE 2.3

- (1) **Multiples of 4 are:** 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48
Multiples of 7 are: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84
Multiples of 9 are: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108
 (2) **Factors of 16 are:** 1, 2, 4, 8, 16 **Factors of 26 are:** 1, 2, 13, 26
Factors of 45 are: 1, 3, 5, 9, 15, 45
 (3) 14, 28, 35, 56, 84 (4) 24, 32, 40, 48, 56, 64, 72, 80, 88
 (5) Factors of 50 are: 1, 2, 5, 10, 25, 50

EXERCISE 2.4

- (1) (i) 1, 2, 3, 4, 6, 12 (ii) 1, 2, 4, 5, 10, 20 (iii) 1, 5, 25
 (iv) 1, 2, 4, 11, 22, 44 (v) 1, 2, 4, 8, 16, 32, 64 (vi) 1, 2, 4, 7, 14, 28
 (2) (i) $2 \times 2 \times 3 \times 3$ (ii) $2 \times 3 \times 3 \times 3$ (iii) $3 \times 3 \times 3$
 (iv) $5 \times 3 \times 3$ (v) $2 \times 2 \times 2 \times 2 \times 2$ (vi) 2×41
 (3) (i) $3 \times 3 \times 7$ (ii) $3 \times 3 \times 5$ (iii) $2 \times 2 \times 2 \times 3 \times 3$
 (iv) $2 \times 3 \times 3 \times 3$ (v) 2×19 (vi) $2 \times 3 \times 3 \times 5$

ANSWERS

EXERCISE 2.5

- (1) (i) $12 = 1, 2, 3, 4, 6, 12$ (ii) $10 = 1, 2, 5, 10$
 $18 = 1, 2, 3, 6, 9, 18$ $15 = 1, 3, 5, 15$
- (iii) $22 = 1, 2, 11, 22$ (iv) $8 = 1, 2, 4, 8$
 $44 = 1, 2, 4, 11, 22, 44$ $32 = 1, 2, 4, 8, 16, 32$
- (v) $24 = 1, 2, 3, 4, 6, 8, 12, 24$
 $36 = 1, 2, 3, 4, 6, 9, 12, 18, 36$
 $30 = 1, 2, 3, 5, 6, 10, 15, 30$
- (2) (i) 6 (ii) 14 (iii) 15 (iv) 25 (v) 18
 (vi) 32 (vii) 30 (viii) 5
- (3) (i) 12 (ii) 9 (iii) 4 (iv) 12 (v) 2 (vi) 7

EXERCISE 2.6

- (1) 12 students (2) 6 books (3) 22 cm
 (4) 6 cm (5) 4 balloons (6) 16 litres (7) 5 days

EXERCISE 2.7

- (1) (i) 75 (ii) 84 (iii) 20 (iv) 18 (v) 35
 (2) (i) 126 (ii) 180 (iii) 180 (iv) 105 (v) 240
 (vi) 216 (vii) 315 (viii) 90 (ix) 66 (x) 360

EXERCISE 2.8

- (1) Ali will take 4 days and Mari will take 3 days (2) 6 seconds time
 (3) 60 inches (4) 8 (5) 5 days (6) 24 days

REVIEW EXERCISE 2

- (1) (i) b (ii) b (iii) c (iv) c (v) a
 (2) HCF = 3, LCM = 3312 (3) HCF = 1, LCM = 1517
 (4) 15 students (5) 30 friends

ANSWERS

EXERCISE 3.1

(1) **Like Fractions** (i), (iii), (vi), (viii) and (ix) **Unlike Fractions** (ii), (iv), (v) and (vii)

(2) (i) = (ii) > (iii) > (iv) < (v) > (vi) >

(3) (i) $\frac{5}{7}$, $\frac{6}{7}$, $\frac{9}{7}$ (ii) $\frac{5}{9}$, $\frac{2}{3}$, $\frac{5}{6}$ (iii) $\frac{4}{15}$, $\frac{1}{3}$, $\frac{2}{5}$

(iv) $\frac{1}{6}$, $\frac{5}{12}$, $\frac{1}{2}$, $\frac{3}{4}$ (v) $\frac{9}{8}$, $\frac{5}{4}$, $\frac{13}{6}$, $\frac{7}{2}$ (vi) $\frac{5}{12}$, $\frac{7}{6}$, $\frac{7}{4}$, $\frac{5}{2}$

(4) (i) $\frac{3}{4}$ (ii) $\frac{2}{3}$ (iii) $\frac{3}{5}$ (iv) $\frac{1}{3}$ (v) $\frac{1}{2}$

EXERCISE 3.2

(1) **Unit Fractions** (i), (ii), (iv), (viii) and (ix), **Proper Fractions** (i), (ii), (iv), (v), (viii) and (ix) **Improper Fractions** (iii) and (vii), **Mixed Fractions** (vi) and (x)

(2) (i) $6\frac{1}{7}$ (ii) $7\frac{1}{4}$ (iii) $8\frac{2}{9}$ (iv) $2\frac{2}{6}$

(v) $5\frac{3}{5}$ (vi) $22\frac{1}{3}$ (vii) $7\frac{8}{11}$ (viii) $17\frac{1}{5}$

(3) (i) $\frac{43}{10}$ (ii) $\frac{17}{3}$ (iii) $\frac{13}{2}$ (iv) $\frac{23}{7}$

(v) $\frac{33}{4}$ (vi) $\frac{24}{13}$ (vii) $\frac{63}{8}$ (viii) $\frac{20}{7}$

EXERCISE 3.3

(1) (i) $1\frac{8}{9}$ (ii) $1\frac{2}{3}$ (iii) $1\frac{4}{5}$ (iv) $1\frac{11}{12}$ (v) $8\frac{5}{6}$

(vi) $8\frac{1}{12}$ (vii) $3\frac{11}{30}$ (viii) $3\frac{24}{35}$

(2) (i) $\frac{1}{3}$ (ii) $\frac{1}{8}$ (iii) $\frac{7}{15}$ (iv) $2\frac{1}{20}$ (v) $6\frac{3}{10}$

(vi) $\frac{17}{30}$

(3) (i) $\frac{7}{4}$ (ii) $\frac{4}{7}$, $\frac{2}{9}$ (iii) $\frac{1}{5}$ (iv) $\frac{2}{7}$, $\frac{3}{10}$

ANSWERS

EXERCISE 3.4

- (1) (i) $7\frac{1}{2}$ (ii) 3 (iii) $9\frac{9}{14}$
 (2) (i) $\frac{1}{5}$ (ii) $\frac{8}{9}$ (iii) $\frac{5}{21}$ (iv) $\frac{5}{12}$
 (v) $\frac{7}{16}$ (vi) $\frac{3}{8}$ (vii) 24 (viii) $27\frac{1}{2}$ (ix) 33

EXERCISE 3.5

- (i) $\frac{1}{4}$ (ii) 3 (iii) 3 (iv) $\frac{4}{7}$ (v) $1\frac{1}{8}$
 (vi) $\frac{3}{5}$ (vii) $2\frac{4}{7}$ (viii) $\frac{3}{10}$ (ix) $5\frac{5}{6}$ (x) $4\frac{2}{3}$
 (xi) $4\frac{1}{5}$ (xii) $\frac{170}{249}$ (xiii) 9 (xiv) $\frac{1}{12}$ (xv) $\frac{4}{15}$

EXERCISE 3.6

- (1) $11\frac{3}{4}$ litres of milk (2) $\frac{7}{30}$ cm long (3) 6 years old
 (4) $2\frac{2}{5}$ metres (5) 27 pieces (6) $37\frac{1}{2}$ bags

REVIEW EXERCISE 3

- (1) (i) $8\frac{2}{8}$ (ii) $5\frac{2}{3}$ (iii) $4\frac{3}{5}$ (iv) 8
 (2) (i) $\frac{44}{5}$ (ii) $\frac{29}{8}$ (iii) $\frac{52}{7}$ (iv) $\frac{11}{2}$
 (3) (i) $\frac{3}{8}, \frac{5}{8}, \frac{9}{8}$ (ii) $\frac{1}{4}, \frac{2}{3}, \frac{5}{6}$ (iii) $\frac{7}{6}, \frac{6}{4}, \frac{5}{2}, \frac{8}{3}$
 (iv) $\frac{3}{11}, \frac{2}{7}, \frac{5}{11}, \frac{4}{5}$
 (4) (i) $\frac{13}{6}, \frac{11}{6}, \frac{7}{6}$ (ii) $\frac{5}{6}, \frac{2}{3}, \frac{3}{5}$ (iii) $\frac{4}{15}, \frac{3}{10}, \frac{9}{20}, \frac{4}{5}$
 (iv) $\frac{4}{3}, \frac{5}{7}, \frac{3}{5}, \frac{1}{3}$
 (5) (i) $\frac{5}{7}$ (ii) $\frac{6}{11}$ (iii) $\frac{5}{2}$ (iv) $\frac{7}{2}$

ANSWERS

- (6) (i) $1\frac{1}{10}$ (ii) $2\frac{1}{9}$ (iii) $1\frac{9}{10}$ (iv) $5\frac{1}{6}$
 (7) (i) $\frac{1}{4}$ (ii) $\frac{2}{5}$ (iii) $1\frac{1}{6}$ (iv) $\frac{1}{6}$
 (8) (i) 6 (ii) $3\frac{3}{4}$ (iii) $\frac{336}{735}$ (iv) $2\frac{2}{5}$
 (9) (i) $\frac{1}{30}$ (ii) $1\frac{1}{5}$ (iii) $4\frac{2}{5}$ (iv) $4\frac{4}{5}$
 (10) $2\frac{5}{6}$ km (11) 8 balloons (12) 5 dresses (13) 133 litres

EXERCISE 4.1

- (2) (i) One (ii) Three (iii) Two
 (iv) Two (v) Three (vi) One
 (3) (i) 4 hundredths = $\frac{4}{100} = 0.04$ (ii) 8 tenths = $\frac{8}{10} = 0.8$
 (iii) 2 thousandths = $\frac{2}{1000} = 0.002$ (iv) 1 hundred = 100
 2 hundredths = $\frac{2}{100} = 0.02$
 (v) 4 tens = 40 (vi) 3 tens = 30
 9 hundredths = $\frac{9}{100} = 0.09$ 7 hundredths = $\frac{7}{100} = 0.07$
 (4) (i) Place value of 5 = 5 hundreds = 500
 Place value of 7 = 7 tens = 70
 Place value of 5 = 5 ones = 5
 Place value of 1 = 1 tenths = 0.1
 (ii) Place value of 0 = 0 ones = 0
 Place value of 5 = 5 tenths = 0.5
 Place value of 3 = 3 hundredth = 0.03
 Place value of 4 = 4 thousandths = 0.004
 (iii) Place value of 1 = 1 tens = 10
 Place value of 8 = 8 ones = 8
 Place value of 9 = 9 tenths = 0.9
 Place value of 7 = 7 hundredth = 0.07
 Place value of 9 = 9 ones = 9
 Place value of 2 = 2 hundredth = 0.02
 (iv) Place value of 9 = 9 ones = 9
 Place value of 0 = 0 tenths = 0.0
 Place value of 2 = 2 hundredth = 0.02
 (v) Place value of 7 = 7 tens = 70
 Place value of 5 = 5 ones = 5
 Place value of 8 = 8 tenths = 0.8
 Place value of 4 = 4 hundredth = 0.04
 (vi) Place value of 5 = 5 tens = 50
 Place value of 1 = 1 ones = 1
 Place value of 1 = 1 tenths = 0.1
 Place value of 5 = 5 hundredth = 0.05

EXERCISE 4.2

- (1) (i) 0.01 (ii) 0.5 (iii) 0.23 (iv) 0.431 (v) 2.31
 (vi) 1.7 (vii) 0.029 (viii) 5.72

A N S W E R S

- (2) (i) 0.28 (ii) 10.5 (iii) 4.5 (iv) 0.75 (v) 0.26
(vi) 0.375 (vii) 0.76 (viii) 0.6

- (3) (i) $\frac{3}{10}$ (ii) $1\frac{7}{20}$ (iii) $12\frac{37}{100}$ (iv) $\frac{54}{125}$ (v) $25\frac{1}{2}$
(vi) $2\frac{8}{25}$ (vii) $\frac{9}{20}$ (viii) $135\frac{1}{5}$

EXERCISE 4.3

- (1) (i) 0.77 (ii) 0.67 (iii) 4.76 (iv) 35.6 (v) 1.77
(vi) 41.07 (vii) 1.6 (viii) 22.6 (ix) 81.011 (x) 123.726
(2) (i) 0.6 (ii) 0.23 (iii) 7.32 (iv) 0.67 (v) 2.98
(vi) 1.78 (vii) 8.99 (viii) 14.95 (ix) 32.086 (x) 12.036

EXERCISE 4.4

- (1) 63 (2) 2.5 (3) 173.2 (4) 32.7 (5) 2235.42
(6) 1265 (7) 1872 (8) 18.9 (9) 21.25 (10) 102.7
(11) 237.93 (12) 20.174 (13) 130.08 (14) 34.79 (15) 158.4

EXERCISE 4.5

- (1) 2.85 (2) 0.36 (3) 2.34 (4) 3.22 (5) 0.52
(6) 2.55 (7) 1.57 (8) 0.36 (9) 2.81 (10) 11.41
(11) 13.45 (12) 5.39 (13) 48.1 (14) 3.37 (15) 8.01

EXERCISE 4.6

- (1) Rs 851.05 (2) 21.52 kg (3) Rs 67.25
(4) Rs. 6.07 (5) Rs. 21.3 (6) 176.8 kg
(7) 120 kg (8) 2.75 m (9) 11.75 kg

REVIEW EXERCISE 4

- (1) (i) 5 thousandths = $\frac{5}{1000} = 0.005$ (ii) 1 tens = 10
9 tenths = $\frac{9}{10} = 0.9$
(iii) 7 hundredths = $\frac{7}{100} = 0.07$

ANSWERS

- (2) (i) 3.21 (ii) 0.175 (iii) 4.75 (iv) 3.375
 (3) (i) $1\frac{27}{50}$ (ii) $\frac{7}{20}$ (iii) $13\frac{7}{10}$ (iv) $\frac{69}{200}$
 (4) (i) 4.99 (ii) 0.19 (iii) 40.97 (iv) 1.05
 (5) (i) 32.5 (ii) 175.5 (iii) 3525.2 (iv) 82.24
 (6) (i) 1.57 (ii) 0.36 (iii) 2.87
 (7) 208 m and 12.5 m (8) 13.5 litres

EXERCISE 5.1

- (1) (i) 5000 m (ii) 14000 m (iii) 20000 m
 (2) (i) 1700 cm (ii) 3200 cm (iii) 5400 cm
 (3) (i) 150 mm (ii) 190 mm (iii) 300 mm
 (4) (i) 400 cm, 4000 mm (ii) 1000 cm, 10000 mm
 (iii) 3500 cm, 35000 mm (iv) 6400 cm, 64000 mm
 (v) 8300 cm, 83000 mm (vi) 9800 cm, 98000 mm

EXERCISE 5.2

- (1) (i) 13800 m (ii) 38 km 360 m (iii) 81 km 822 m
 (iv) 59 km 89 m (v) 168 m 49 cm (vi) 110 cm 8 mm
 (vii) 118 m 30 cm (viii) 276 km 495 m (ix) 357 km 919 m
 (2) (i) 1950 m (ii) 3m 59 cm (iii) 2 m 60 cm
 (iv) 557 m 40 cm (v) 50 km 750 m (vi) 51 km 748 m
 (vii) 194 m 11 cm (viii) 46 cm 9 mm (ix) 26 cm 8 mm

EXERCISE 5.3

- (1) (i) 2cm (b) (ii) 30 m (c) (iii) 165 km (d)
 (2) 1 m 48 cm (3) 1157 m (4) 79 cm (5) 1 cm 17 mm
 (6) 350 m (7) 9km 755m (8) 150 cm (9) 130 m

EXERCISE 5.4

- (1) (i) 16241 g (ii) 11 kg 300 g (iii) 17kg 121g
 (iv) 7kg 117g (v) 89 kg 765 g
 (2) (i) 3 kg 750 g (ii) 3kg 53g (iii) 4502 g
 (iv) 23kg 220g (v) 5kg 968g

ANSWERS

EXERCISE 5.5

- (1) (i) (a) (ii) (c) (iii) (b) (iv) 5 kg (a)
 (2) 162 kg 100 gm (3) 34 kg 750 g (4) 22 kg 550 g (5) 1143 kg 250 g

EXERCISE 5.6

- (1) (i) 17 l 920 ml (ii) 10815 ml (iii) 54 l 676 ml
 (iv) 2220 ml (v) 58 l 532 ml
 (2) (i) 106 l (ii) 44 l 185 ml (iii) 265 ml
 (iv) 2 l 945 ml (v) 21 l 437 ml

EXERCISE 5.7

- (1) (i) 4 l (ii) 1 l (iii) 500 ml
 (iv) 4 l (v) 140 ml (vi) 5 ml
 (2) 9 ml (3) 298 l water needed (4) 150 l
 (5) 236 l 125 ml (6) 499 l 740 ml (7) 3 l 85 ml (8) 19 l 250 ml

EXERCISE 5.8

- (1) (i) 60 months (ii) 102 months (iii) 57 months
 (iv) 122 months (v) 188 months (vi) 250 months
 (2) (i) 90 days (ii) 102 days (iii) 260 days
 (iv) 145 days (v) 328 days (vi) 75 days
 (3) (i) 56 days (ii) 178 days (iii) 125 days
 (iv) 210 days (v) 250 days (vi) 289 days

EXERCISE 5.9

- (1) (i) 76 minutes 18 seconds (ii) 85 minutes 59 seconds
 (iii) 59 hours 30 minutes (iv) 59 hours 35 minutes
 (2) (i) 11 minutes 10 seconds (ii) 20 minutes 10 seconds
 (iii) 13 hours 34 minutes (iv) 31 hours 20 minutes

EXERCISE 5.10

- (1) 48 minutes 55 seconds (2) 7 hours 45 minutes
 (3) 10 minutes 42 seconds (4) 2 hours 25 minutes
 (5) 14 minutes 24 seconds

ANSWERS

REVIEW EXERCISE 5

- (1) (i) (b) (ii) (a) (iii) (c) (iv) (b)
 (3) (i) 6 km (ii) 3.5 km (4) (i) 15000 m (ii) 3000 m
 (5) (i) 48km 127m (ii) 87 m 41 cm (iii) 26 l 500 ml (iv) 83 kg 340 g
 (6) (i) 36 km 7 m (ii) 28 m 53 cm (iii) 8 l 250 ml (iv) 41 kg 313 g

EXERCISE 6.1

- (1) (i) 3 cm, 5.6 cm, 3.5 cm, 3.5 cm (ii) 5.4 cm, 5.4 cm, 2.6 cm, 3.3 cm
 (iii) 2.7 cm, 2.7 cm, 3.5 cm, 3.5 cm
 (2) (ii) 7.9 cm (iii) 9.1 cm (iv) 10 cm (v) 10.6 cm

EXERCISE 6.2

- (2) (i) 4.2 cm (ii) 4.7 cm (iii) 4.3 cm

EXERCISE 6.3

- (1) West and East (2) North and South (3) Vertical (4) Horizontal

EXERCISE 6.4

- (1) (i), (iv), (vi) and (viii) are parallel lines
 (ii), (iii), (v) and (vii) are non-parallel lines

EXERCISE 6.6

- (1) (i) Vertex B, Arms $\overrightarrow{BA}$ and $\overrightarrow{BC}$ (ii) Vertex Q, Arms $\overrightarrow{QP}$ and $\overrightarrow{QR}$
 (iii) Vertex W, Arms $\overrightarrow{WX}$ and $\overrightarrow{WY}$
 (2) (i) $\angle AOB$ (ii) $\angle AEF$ (iii) $\angle JKL$

EXERCISE 6.7

- (1) (iii), (iv) and (v) are right angles (2) (i), (ii), (v) and (vi) are right angles

EXERCISE 6.8

- (1) (i) right (ii) acute (iii) obtuse (iv) acute (v) obtuse (vi) right

EXERCISE 6.9

- (1) 30° (2) 40° (3) 40° (4) 28° (5) 43°
 (6) 43° (7) 110° (8) 110° (9) 142°

ANSWERS

EXERCISE 6.10

- (1) (i) 43° acute angle (ii) 43° acute angle (iii) 70° acute angle
(iv) 155° obtuse angle (v) 90° right angle (vi) 155° obtuse angle

REVIEW EXERCISE 6

- (1) (i) Ruler (ii) Set square (iii) Set square
(iv) Protractor (v) Compasses (vi) Dividers
- (9) (i) 3 angles, $\angle ABD$ acute angle
 $\angle CBD$ acute angle
 $\angle ABC$ right angle
(ii) 6 angles, $\angle GDP$ acute angle, $\angle PDE$ acute angle,
 $\angle GDE$ right angle, $\angle CDE$ obtuse angle,
 $\angle CDG$ obtuse angle, $\angle CDP$ obtuse angle
- (10) (i) 2 Squares, ABED and BCFE
(ii) 5 squares, ABHG, BCIH, HIFE, GHED, ACFD (11) (i) 3 triangles

EXERCISE 7.1

- (1) (i) 5 students (ii) Pink (iii) Yellow
(iv) 20 students (v) 15 students (vi) 6 colours (vii) 95 students
- (2) (i) 20 students (ii) Cricket (iii) 25 students
(iv) 4 games (v) Hockey (vi) 35 students (vii) 110 students

EXERCISE 7.2

- (1) (i) Team D (ii) Team E (iii) No team
(iv) 10 minutes (v) 40 minutes (vi) 15 minutes
- (2) (i) Thursday (ii) 10 acres (iii) 7 days (iv) 10 acres
(v) Monday and Friday (vi) 65 acres
(vii) 15 acres, 5 acres, 10 acres (viii) Monday, Friday; Tuesday, Wednesday, Saturday, Sunday.
- (3) (i) Math and Islamiyat (ii) 360 marks (iii) 70 marks
(iv) 80 marks (v) Science (vi) 70 marks

REVIEW EXERCISE 7

- (1) (i) 80 rupees (ii) 120 rupees (iii) No (iv) 20 rupees (v) 6 litres
- (2) (i) 3000 rupees (ii) 4000 rupees (iii) June
(iv) January (v) 18000 rupees (vi) 1500 rupees
- (3) (i) 2 cm (ii) 6 cm (iii) 2 cm (iv) 6 cm (v) 4^{th} and 5^{th} days (vi) 5 times